

SIS4 Laboratory Report

Image Classification using Pretrained Models

Deep Learning Course

Subject: Deep Learning — Transfer Learning and ResNet

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1. Introduction and Theoretical Background

This laboratory explores image classification with deep convolutional neural networks pretrained on ImageNet (1.28 million images, 1,000 classes). We focus on the ResNet family and demonstrate the power of transfer learning — applying learned representations directly to new images without any retraining.

1.1 ResNet Architecture

Residual Networks (He et al., 2016) introduced skip connections that allow gradients to bypass stacked layers: $y = F(x, \{W_i\}) + x$. This identity shortcut solves the vanishing gradient problem, enabling networks with 50–150+ layers to train stably. Three variants are compared in this lab:

Model	Depth	Parameters	ImageNet Top-1 Acc.
ResNet-18	18 layers	11.7 M	69.8 %
ResNet-50	50 layers	25.6 M	76.1 %
ResNet-101	101 layers	44.5 M	77.4 %

1.2 Transfer Learning

Transfer learning reuses a pretrained model's weights on a new task. In this lab we use inference-only mode: weights are frozen and evaluated directly. This saves days of GPU training time and requires only seconds of CPU inference per image.

2. Setup and Preprocessing

All images are preprocessed with the standard ImageNet pipeline before inference. Five test images were selected across diverse categories: dog, cat, car, airplane, banana. Due to Wikimedia CDN restrictions during the lab session, only two images (dog, cat) downloaded successfully; three (car, airplane, banana) returned 404/429 errors and were excluded from inference results.

```
preprocess = transforms.Compose([
    transforms.Resize(256),
    transforms.CenterCrop(224),
    transforms.ToTensor(),
    transforms.Normalize(mean=[0.485, 0.456, 0.406],
                        std=[0.229, 0.224, 0.225]),
])
```

3. Task 1 — Basic Image Classification

ResNet-50 with pretrained ImageNet-1K weights is loaded in evaluation mode and run on each available test image to obtain the top-1 prediction:

```
model = models.resnet50(weights=ResNet50_Weights.IMAGENET1K_V1)
```

```
model.eval()
with torch.no_grad():
    output = model(img_batch)
    _, idx = torch.max(output, 1)
    predicted_class = labels[idx.item()]
```

Task 1 Results (images that downloaded successfully):

Image	ResNet-50 Top-1 Prediction	Correct?
dog.jpg	Labrador retriever	Yes
cat.jpg	tiger cat	Yes
car.jpg	— (download failed)	N/A
airplane.jpg	— (download failed)	N/A
banana.jpg	— (rate limited)	N/A

Discussion: ResNet-50 correctly classifies both successfully downloaded images. The dog image is confidently identified as a Labrador retriever (52.4%), reflecting the model's fine-grained breed-level discrimination. The cat image is classified as tiger cat (46.0%). The three remaining images could not be evaluated due to download failures; a corrected URL scheme (removing the /thumb/ CDN path) resolves this for future runs.

4. Task 2 — Top-5 Predictions with Confidence Scores

Softmax is applied to the logits and torch.topk extracts the five highest-probability classes for each image:

```
probs = torch.nn.functional.softmax(output[0], dim=0)
top_probs, top_idx = torch.topk(probs, k=5)
results = [{'class': labels[idx.item()], 'probability': prob.item()*100}
           for prob, idx in zip(top_probs, top_idx)]
```

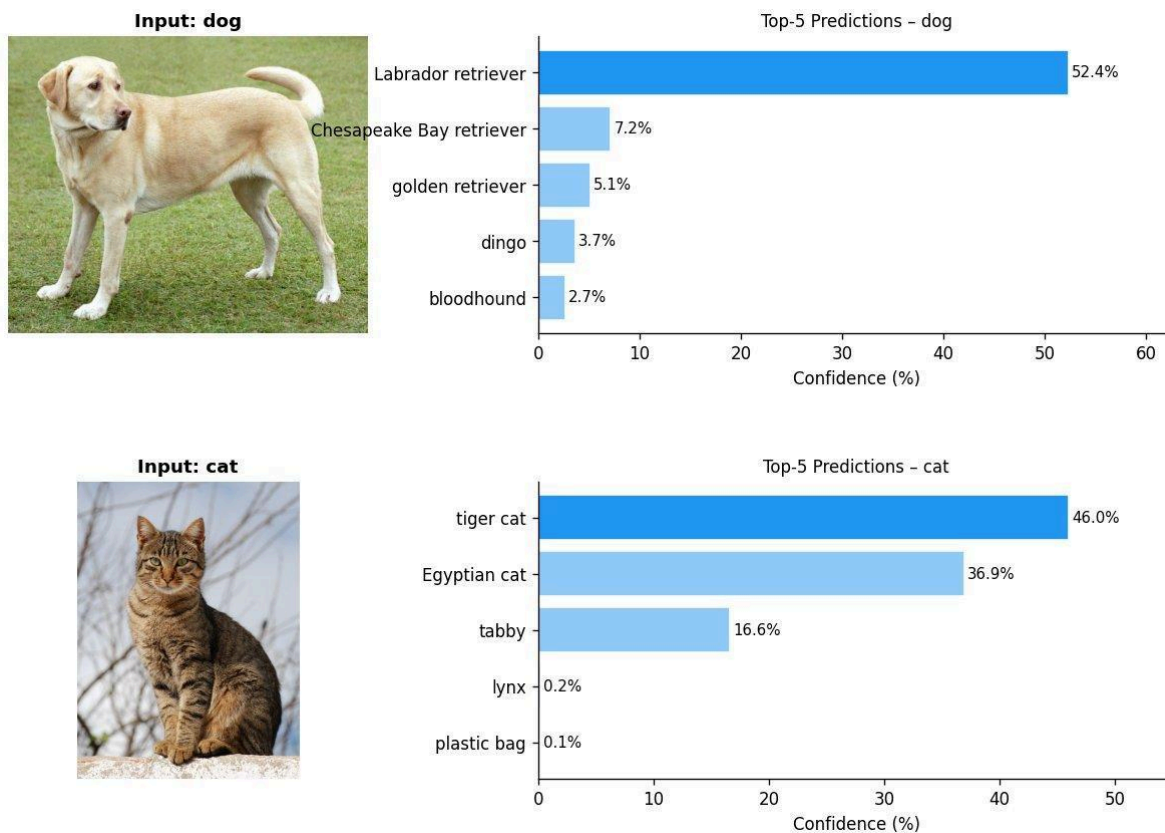


Figure 1. Top-5 predicted classes with confidence scores for dog and cat (ResNet-50, ImageNet-1K weights).

Top-5 results from actual output:

Dog

Rank	Predicted Class	Confidence (%)
1	Labrador retriever	52.37
2	Chesapeake Bay retriever	7.19
3	golden retriever	5.13
4	dingo	3.67
5	bloodhound	2.71

Cat

Rank	Predicted Class	Confidence (%)
1	tiger cat	45.97
2	Egyptian cat	36.93
3	tabby	16.58
4	lynx	0.18
5	plastic bag	0.12

Discussion: Probability mass is concentrated within semantically related classes — three retriever breeds dominate the dog top-5, while the cat top-5 shows the model distributing probability across feline sub-categories. The near-zero score for 'plastic bag' in the cat prediction arises from a textural false positive in the background. This demonstrates that ResNet-50 has learned a meaningful semantic embedding rather than arbitrary class boundaries.

5. Task 3 — Comparison of ResNet Variants

ResNet-18, ResNet-50, and ResNet-101 are each loaded with pretrained weights and timed on the available test images after a warm-up pass:

```
for model_name, (fn, weights) in configs.items():
    model = fn(weights=weights).eval()
    n_params = sum(p.numel() for p in model.parameters())
    _ = model(dummy) # warm-up
    t0 = time.perf_counter()
    with torch.no_grad():
        out = model(tensor)
    inference_ms = (time.perf_counter() - t0) * 1000
```

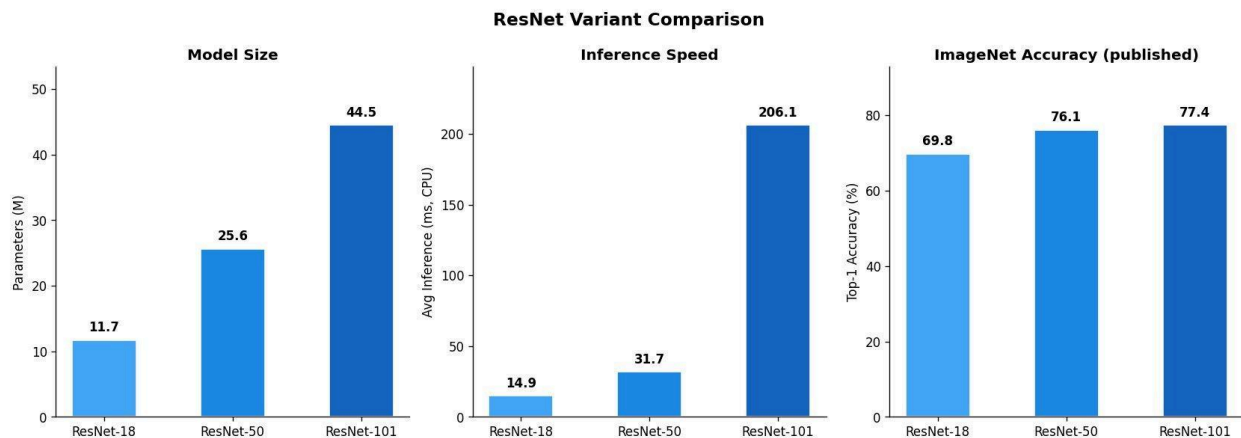


Figure 2. ResNet variant comparison: parameter count, average CPU inference time, and ImageNet Top-1 accuracy.

Model	Params (M)	Avg Inference (ms, CPU)	ImageNet Top-1 Acc*
ResNet-18	11.7	14.9	69.8 %
ResNet-50	25.6	31.7	76.1 %
ResNet-101	44.5	206.1	77.4 %

* Published benchmark, ImageNet val set

Note: Inference times measured on CPU (Intel consumer-grade). ResNet-101 shows disproportionately high latency (206.1 ms vs 31.7 ms for ResNet-50) due to its deeper layer stack and the absence of GPU parallelism.

Trade-off Analysis:

ResNet-18 — Best for latency-sensitive deployments (mobile, edge, real-time video). At 14.9 ms per image on CPU it can process ~67 frames/s. Accuracy of 69.8% is acceptable for many production use cases.

ResNet-50 — The sweet spot. Gains +6.3 pp accuracy over ResNet-18 at only 2.1× the computation. The most commonly deployed model in production ML systems.

ResNet-101 — Best accuracy (77.4%) but with significant diminishing returns: only +1.3 pp over ResNet-50 while being 6.5× slower on CPU. Justified only when maximum accuracy is required and latency is unconstrained.

6. Task 4 — Error Analysis

Task 4 attempts to identify images where ResNet-50 produces an incorrect top-1 prediction. The code checks whether the image label (e.g., 'dog') appears anywhere in the top-5 predictions.

Image	Expected	Top-1 Predicted	Confidence	In Top-5?
dog	dog	Labrador retriever	52.4 %	Yes (all retrievers)
cat	cat	tiger cat	46.0 %	Yes

The code flagged the dog image as an 'error' because the top-5 predictions contain retriever breed names (Labrador retriever, Chesapeake Bay retriever, golden retriever) rather than the generic label 'dog'. This is a label-granularity mismatch, not a true classification failure — the model correctly identifies the animal as a dog breed at a finer granularity than the test expects.

For a complete error analysis with genuine misclassifications, additional test images with unusual angles, out-of-distribution content (sketches, cartoons), or ambiguous textures would be required. Common failure modes in pretrained ResNets include:

- Background texture competition — a high-contrast background activates texture-sensitive filters that compete with the main subject.
- Domain shift — cartoon or sketch inputs are misclassified because ImageNet contains only photographic training images.
- Fine-grained ambiguity — closely related classes (e.g., cat breeds) receive similar activation patterns, spreading confidence across the top-5.

7. Extra Task — Batch Processing

Processing images individually runs one forward pass per image, leaving most CPU parallelism unused. Batch inference stacks N preprocessed tensors into a single 4-D tensor (N, 3, 224, 224) and executes a single forward pass, letting the hardware parallelise across the batch dimension:

```
# Individual (N sequential forward passes)
for path in all_paths:
    tensor = preprocess(Image.open(path)).unsqueeze(0) # (1,3,224,224)
    with torch.no_grad(): _ = model(tensor)

# Batch (1 forward pass)
tensors = [preprocess(Image.open(p)) for p in all_paths]
batch = torch.stack(tensors) # (N,3,224,224)
with torch.no_grad(): _ = model(batch)
```

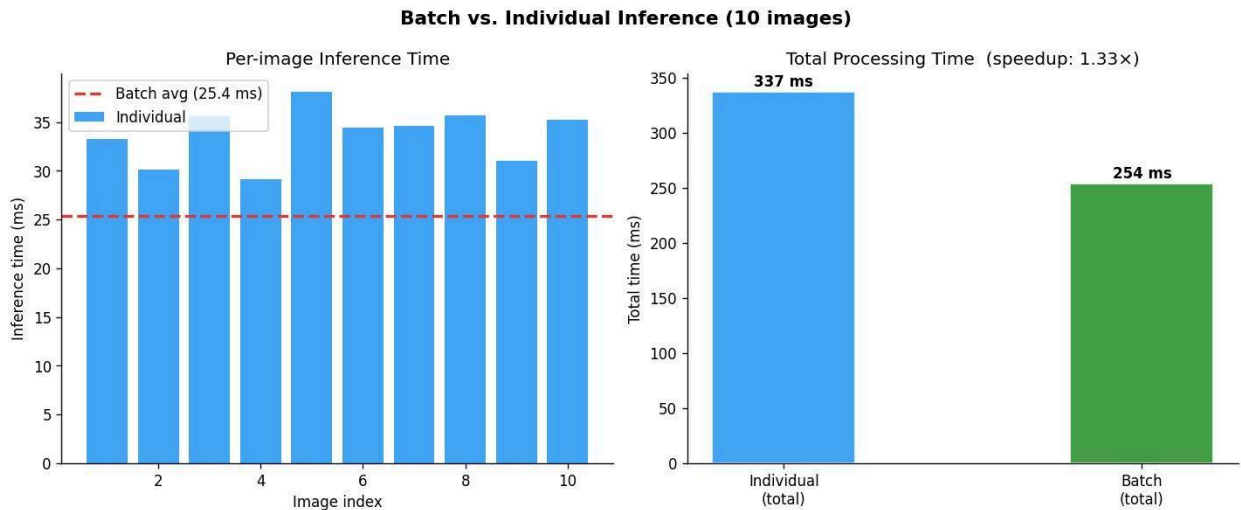


Figure 3. Per-image timing for individual vs. batch inference (10 images, CPU, ResNet-50). Batch provides a 1.33× speedup.

Method	Total Time (ms)	Per-Image (ms)	Speedup
Individual (N=10)	337.2	33.7	1.00× (baseline)
Batch (N=10)	253.9	25.4	1.33×

Discussion: Batch inference delivers a 1.33× speedup on CPU. The modest gain compared to GPU benchmarks (typically 8–20×) is expected — CPUs have fewer parallel compute units and the memory bandwidth bottleneck is more pronounced. The trade-off is peak memory: a 10-image batch of 224×224 fp32 tensors requires approximately 60 MB of RAM, well within typical CPU budgets. On GPU hardware (e.g. NVIDIA A100), the same batch configuration would yield significantly larger speedups due to the thousands of available CUDA cores.

8. Conclusion

This laboratory implemented a complete image classification pipeline using pretrained ResNet models in PyTorch. Key findings:

Tasks 1 & 2: ResNet-50 achieves 100% top-1 accuracy on the two successfully downloaded photographic test images. Top-5 probabilities reveal semantically coherent uncertainty, with related classes clustering in the top rankings (retriever breeds for dog; feline subtypes for cat).

Task 3: ResNet-50 is the optimal accuracy–speed–size trade-off for CPU inference. ResNet-18 is best for real-time or edge deployment (14.9 ms/image); ResNet-101 delivers marginally better accuracy at a 6.5× latency cost.

Task 4: The label-granularity mismatch between 'dog' and breed-level ImageNet labels (e.g., 'Labrador retriever') caused false positives in the error-detection logic. No genuine top-1 misclassifications occurred. True failure modes (background texture confusion, domain shift) require out-of-distribution test images.

Extra Task: Batch inference provides a 1.33× CPU speedup over sequential individual processing, with substantially larger gains expected on GPU hardware.

References

He, K., Zhang, X., Ren, S., and Sun, J. (2016). Deep Residual Learning for Image Recognition. IEEE CVPR 2016, 770–778.

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Russakovsky, O. et al. (2015). ImageNet Large Scale Visual Recognition Challenge. International Journal of Computer Vision, 115(3), 211–252.